

■ Predictive Fleet Maintenance Dashboard

Complete Project Documentation & Guide

■ Project Overview

The Predictive Fleet Maintenance Dashboard is a full-stack AI-powered application designed for real-time monitoring and predictive maintenance of fleet tyres.

Key Features:

- ■ Real-time tyre monitoring (PSI, depth, temperature)
- ■ AI-powered predictions using RandomForest models
- ■ Batch processing with 60-second auto-cycling
- ■ Modern responsive UI with glassmorphism design
- ■ CSV upload system with drag-and-drop interface
- ■ Session management for continuous monitoring

■■ Technology Stack

Backend:

- Django 5.2.4 - Web framework
- Django REST Framework 3.16.0 - API development
- SQLite - Database
- scikit-learn 1.7.1 - Machine Learning

- pandas 2.3.1 - Data processing
- numpy 2.3.2 - Numerical computing

Frontend:

- React 18.2.0 - UI framework
- CSS3 - Styling with glassmorphism effects
- JavaScript ES6+ - Modern JavaScript features

■■ Installation & Setup

Prerequisites:

- Python 3.8+
- Node.js 14+
- npm or yarn

Backend Setup:

```
cd backend
python -m venv venv
venv\Scripts\activate # Windows
pip install -r ../requirements.txt
python manage.py migrate
```

Frontend Setup:

```
cd frontend
npm install
```

■ Running the Application

Terminal 1 - Backend:

```
cd backend
python manage.py runserver
URL: http://localhost:8000
```

Terminal 2 - Frontend:

```
cd frontend  
npm start  
URL: http://localhost:3000
```

■ API Documentation

Base URL: <http://localhost:8000/api/>

1. Upload CSV Data

Endpoint: POST [/api/upload/](http://localhost:8000/api/upload/)

Upload CSV file with tyre data for processing

2. Get Batch Data

Endpoint: GET [/api/batch/](http://localhost:8000/api/batch/)

Retrieve processed tyre data for a specific batch

■ Machine Learning Implementation

ML Models:

- RandomForestClassifier - Tyre condition (Good/Average/Bad)
- RandomForestRegressor - Remaining kilometers

■ CSV Data Format

Required columns: tyre_id, psi, depth, temp

Example:

```
tyre_id, psi, depth, temp
```

T1, 95.5, 7.2, 28.3
T2, 88.1, 5.8, 32.1

■ How to Use the Dashboard

1. Start both servers (backend on :8000, frontend on :3000)
2. Open browser to <http://localhost:3000>
3. Click upload button and drag/drop sample_tyre_data.csv
4. Watch real-time tyre monitoring with AI predictions
5. See auto-cycling every 60 seconds

■ Troubleshooting

- Port already in use: Use different ports or kill existing processes
- Module not found: Activate virtual environment and install requirements
- npm install errors: Clear cache and reinstall
- CORS errors: Check CORS settings in Django settings.py

**Built with ❤️■ using Django, React,
and Machine Learning**

This project demonstrates a complete full-stack application with AI integration, making it an excellent learning resource for modern web development!